Revision

Work & Energy

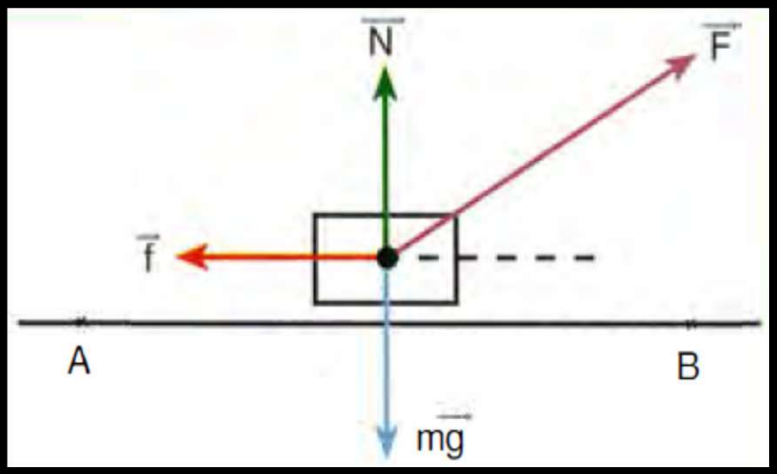
Key Terms

- Work
- Energy
- Kinetic Energy
- Potential Energy

Key Terms

- Mechanical Energy
- Conservation of ME
- Work-Kinetic Energy Theorem
- Power

Work done by a constant force



- Case of a rectilinear displacement $\overrightarrow{AB}$:

$$W_{\vec{F}} = \vec{F} \cdot \overrightarrow{AB} = F \cdot AB \cdot \cos(\alpha)$$

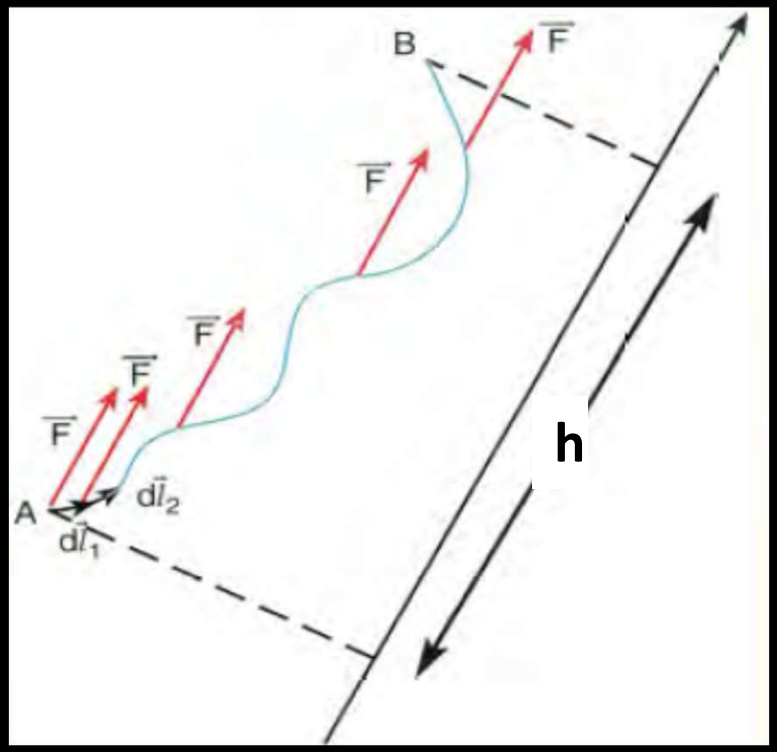
where α is the angle between $\vec{F}$ and $\overrightarrow{AB}$.

- If $0 \leq \alpha < 90° \Rightarrow \cos(\alpha) \geq 0 \Rightarrow W > 0$: motive work
- If $\alpha = 90° \Rightarrow \cos(\alpha) = 0 \Rightarrow W = 0$: no work done by the force
- If $90° < \alpha \leq 180° \Rightarrow \cos(\alpha) < 0 \Rightarrow W < 0$: resistive work

Note

F is measured in (N), distance in (m); work is measured in (N.m) known as joules (J).

Work done by a constant force



- Case of a curvilinear displacement:

$$W_{\vec{F}} = F \cdot AB \cdot \cos\left(\vec{F}, \overrightarrow{AB}\right) = \pm F \cdot h$$

In particular, the work of an object's weight can be found by:

$$W_{\vec{w}} = m \cdot g \cdot (h_i - h_f)$$

Application 1

A skier slides down an inclined plane making $\alpha = 30^\circ$ above the horizontal with a speed v = 54 km/h. w = m.g = 800 N

1) Find the work done by the skier weight during 20 s.
2) Find the magnitude *f* of the frictional force.
3) Calculate the work of this force during 20 s.

Energy

- Energy is the ability to do work.
- So, to do work energy is used. In addition, the object receiving work, receives energy from the agent doing work.
- Both work and energy are measured in joules (J).

Kinetic Energy

Energy due to an object's motion is called kinetic energy

$$KE = \frac{1}{2}mv^2$$

Where:

m is the object's mass; m (kg)

v is the object's speed; v (m/s)

KE is then measured in joules (J)

A 6 kg cat runs after a mouse at 10 m/s. Calculate the cat's kinetic energy.

What is the speed of a 0.145 kg baseball if its kinetic energy is 109 J?

Note for GS

In case of rotational motion, the rotational kinetic energy can be found by:

$$KE_{rot} = \frac{1}{2} I \theta'^2$$

Where:

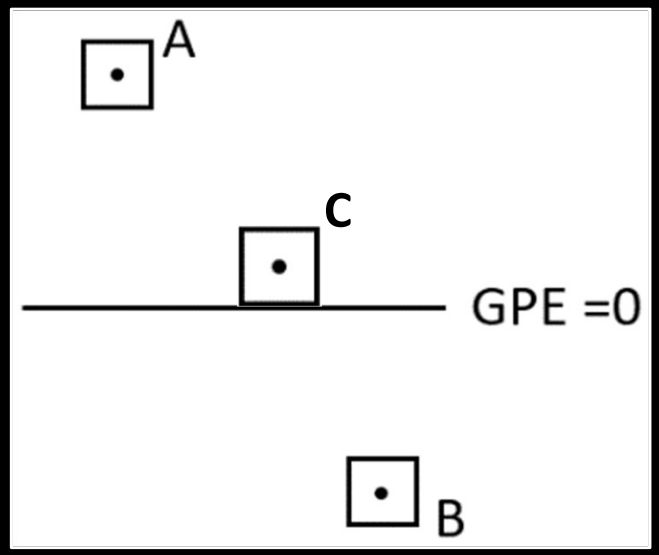
I : is the moment of inertia of the rotating object about the axis of rotation; I(kg.m^2).

θ' : is the angular speed of rotation; θ'(rad/s).

Potential Energy

- Gravitational
- Elastic

Potential Energy



- Gravitational potential energy: it is due to the interaction between the object's mass and the earth

$$GPE = \pm mgh$$

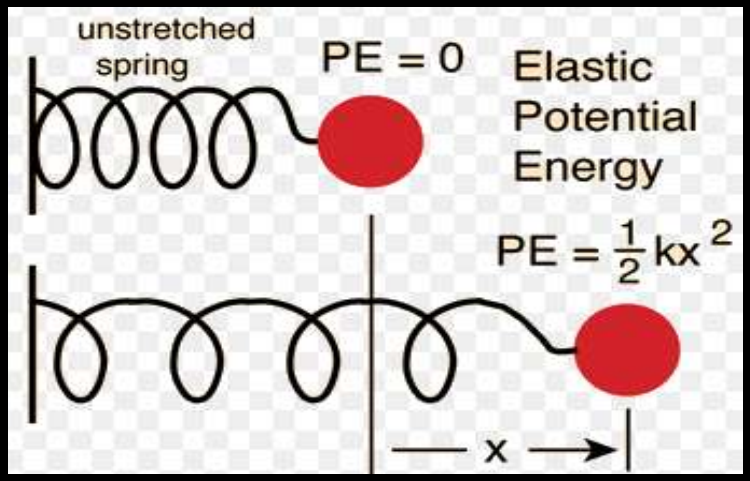
$GPE_A = +mgh_A$ (since A is above the reference)

$GPE_B = -mgh_B$ (since B is below the reference)

$GPE_C = 0$ (since C belongs to the reference of GPE)

Note: Variation of gravitational potential energy: $\Delta GPE = -W_{\vec{w}}$

Potential Energy



- Elastic potential energy: stored in stretched or compressed spring

$$EPE = \frac{1}{2}kx^2$$

where

k *is the spring constant*; k (N/m)

$x = \Delta l = l - l_0;\ \ x\ (m)$

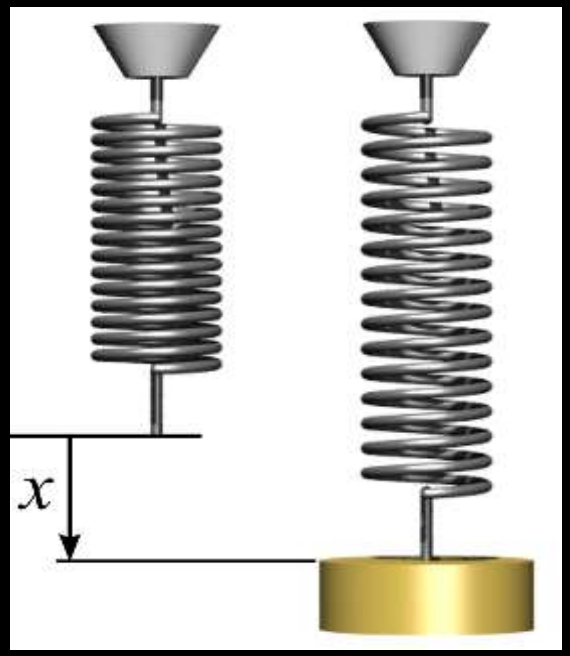
Application 2

A mass m = 100 g is hanging vertically on a spring of original length L_0 = 20 cm and a spring constant k = 20 N/m.

a) Make a free body diagram for m at equilibrium.

b) Find the spring stretched distance. Deduce its new length.

c) Calculate the potential energy of the system (m – earth – spring).



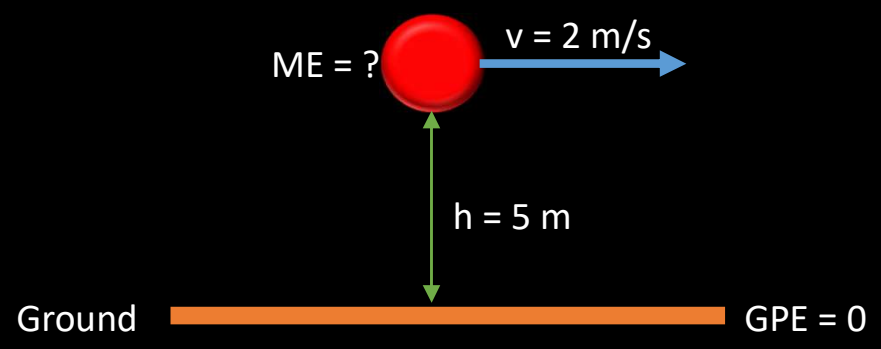
Mechanical Energy

Mechanical energy is the sum of a system's kinetic and potential energy:

$$\mathbf{ME = KE + \Sigma PE = KE + GPE + EPE}$$

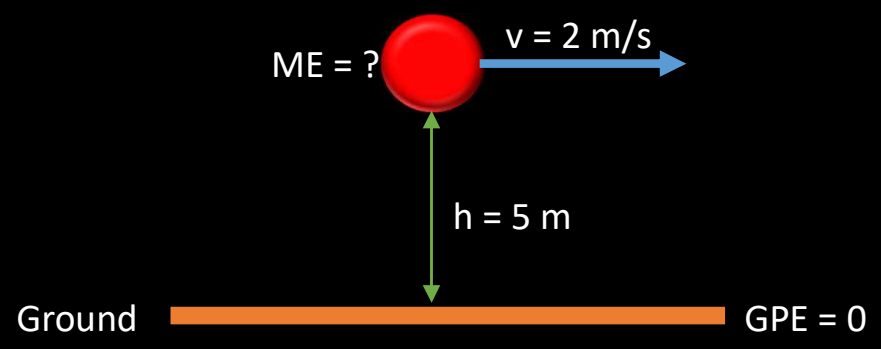
Application 3

A basketball of mass 0.625 kg is moving at 2 m/s at 5m above the ground. Find the basketball mechanical energy.



Solution

A basketball of mass 0.625 kg is moving at 2 m/s at 5m above the ground. Find the basketball mechanical energy.



Solution

$$ME = KE + GPE$$
$$= \frac{1}{2} mv^2 + mgh$$
$$= 0.5 \times 0.625 \times 22 + 0.625 \times 10 \times 5$$
$$= 1.25 + 31.25$$
$$= 32.5 \text{ J}$$

Work-kinetic energy theorem

The total work done on an object is equal to the change in the object's kinetic energy:

$$\Delta \text{KE} = \sum W_{\vec{f}_{ext}}$$

Or

$$w_{tot} = KE_f - KE_i$$

Application 4

Refind the frictional force using the work-KE theorem

A skier slides down an inclined plane making $\alpha = 30^\circ$ above the horizontal with a speed v = 54 km/h. w = m.g = 800 N

1) Find the work done by the skier weight during 20 s.
2) Find the magnitude *f* of the frictional force.
3) Calculate the work of this force during 20 s.

Application 5

(3 methods: work-kinetic energy theorem – Newton's 2nd law ~~– Energy~~)

On a frozen pond, a person kicks a 10 kg sled, giving it an initial speed of 2.2 m/s. How far does the sled move before it stops if the frictional force opposing motion is 10 N.

We saw that

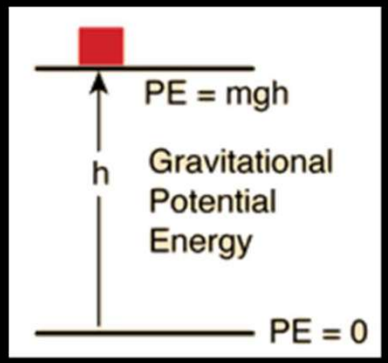
$$KE = \frac{1}{2} mv^2$$

$$\text{kinetic energy} = \frac{1}{2} \times \text{mass} \times (\text{speed})^2$$

Gravitational Potential Energy

As an object is moved against gravity, the work done on the object is stored as gravitational potential energy. As result:

$$PE_g = m.g.h$$

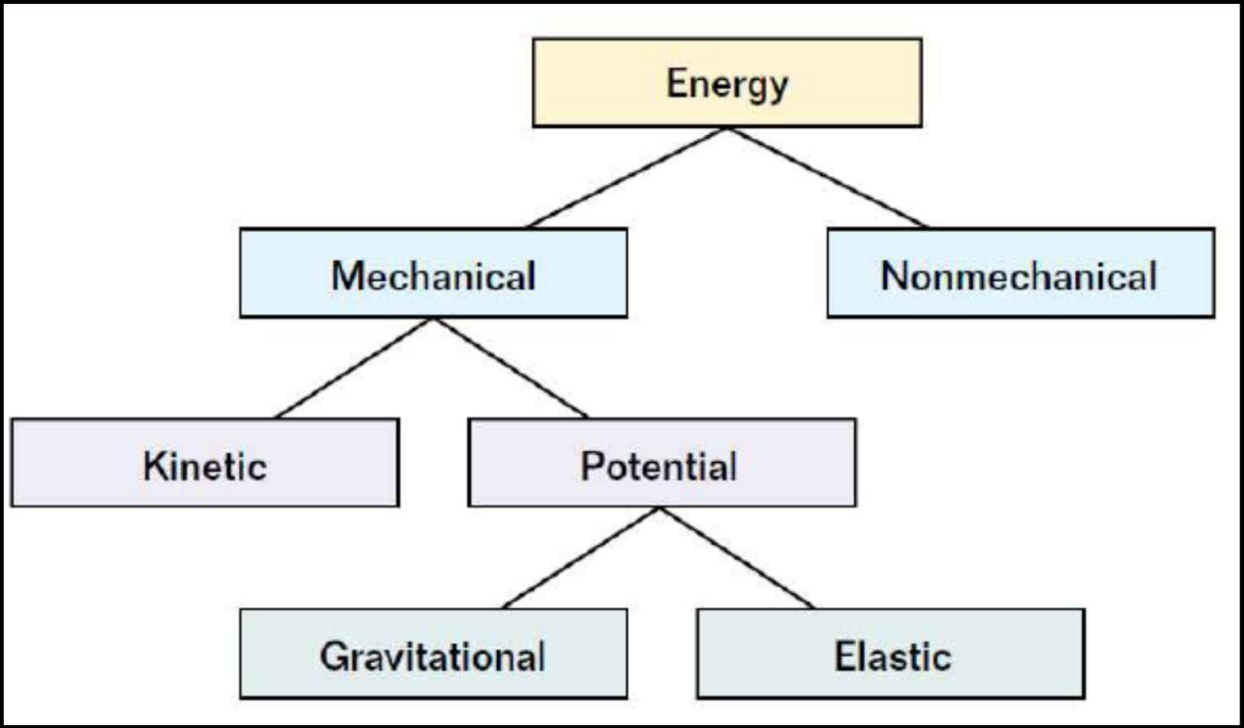


Mechanical Energy

The sum of kinetic energy and of all forms of potential energy is defined as mechanical energy:

$$ME = KE + \Sigma PE$$

We saw that


Energy
Mechanical
Nonmechanical
Kinetic
Potential
Gravitational
Elastic

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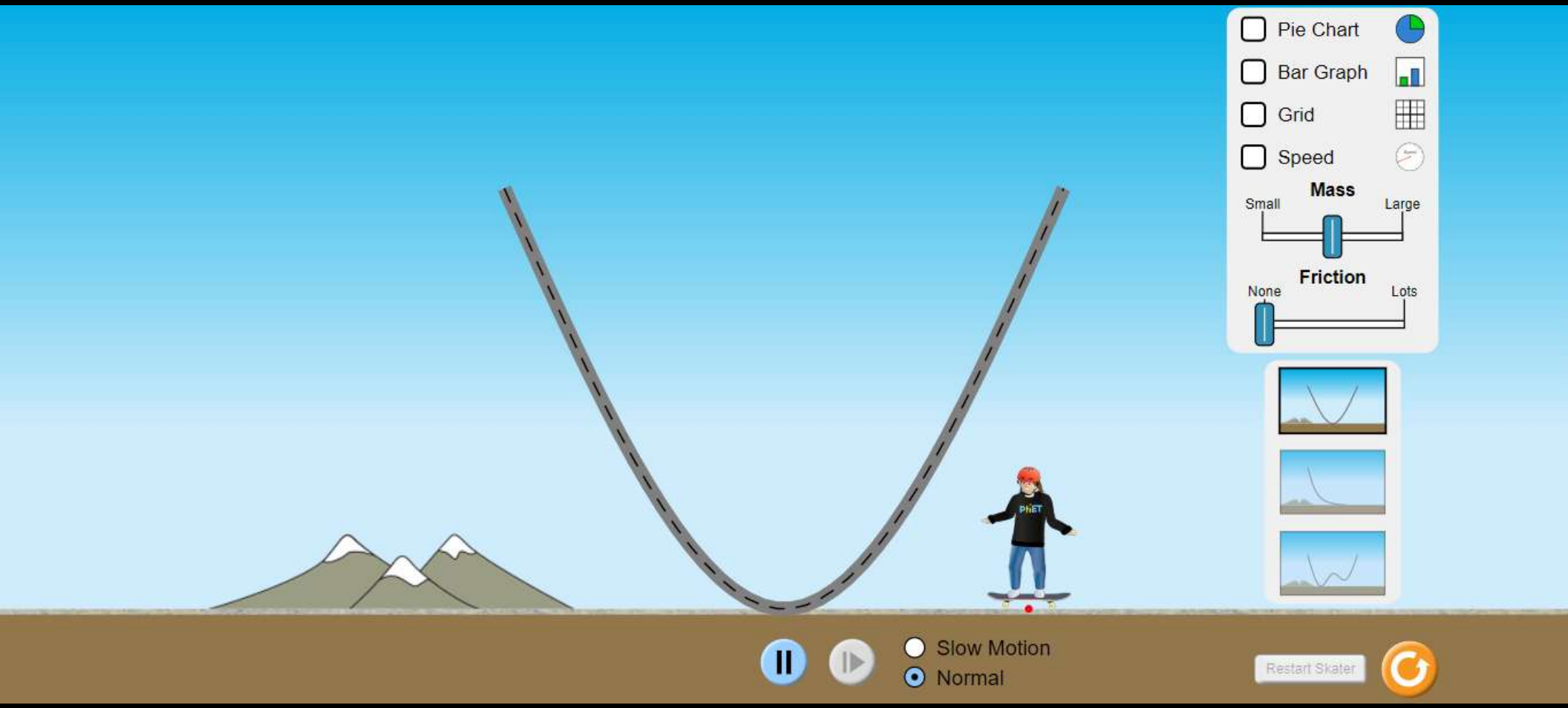
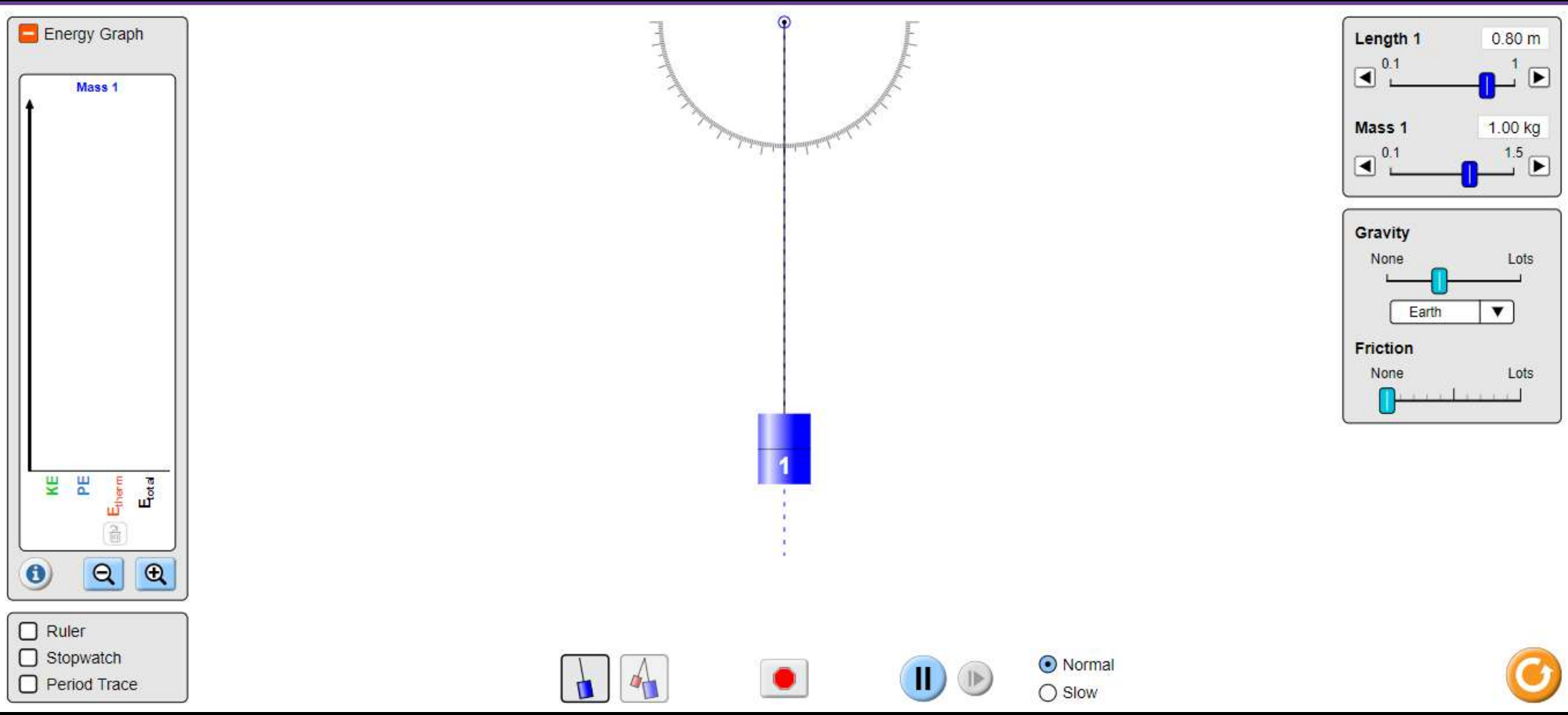
Simulations


Pie Chart
Bar Graph
Grid
Speed
Mass
Small
Large
Friction
None
Lots
Slow Motion
Normal
Restart Skater


Energy Graph
Mass 1
KE
PE
E therm
E total
Ruler
Stopwatch
Period Trace
1
Length 1
0.80 m
0.1
1
Mass 1
1.00 kg
0.1
1.5
Gravity
None
Lots
Earth
Friction
None
Lots
Normal
Slow

Conservation of Mechanical Energy

In **absence of friction** a system's **mechanical energy** will remain unchanged; that is **potential energy** can be converted into **kinetic energy** and vice versa, but their sum keeps same:

final mechanical energy = initial mechanical energy

(in the absence of friction)

Or

$$ME_f = ME_i$$

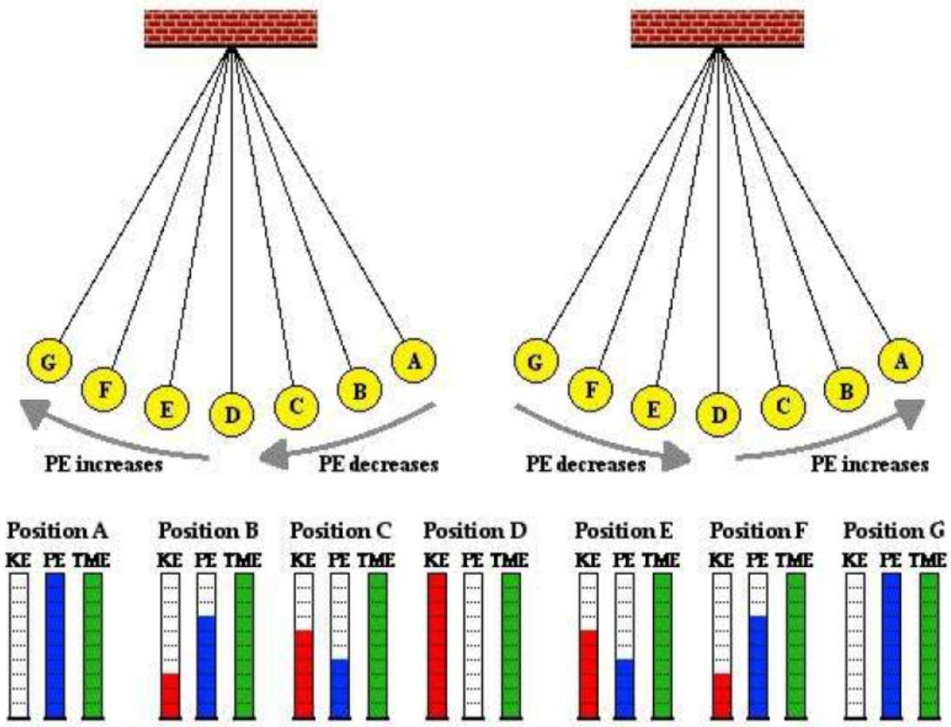
Or

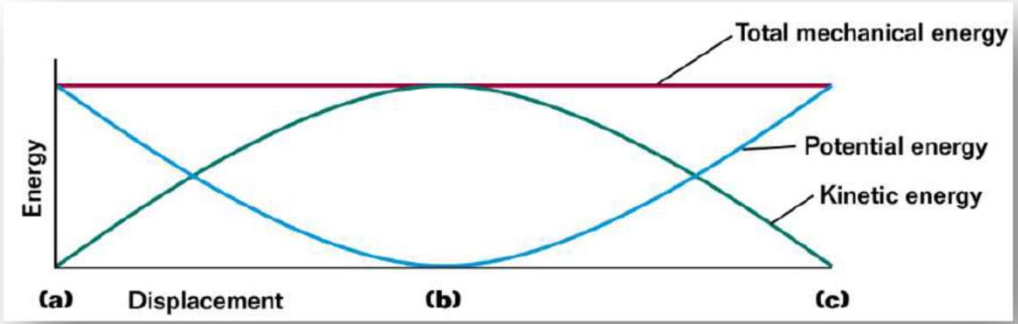
$$KE_f + GPE_f = KE_i + GPE_i$$

Or

$$\frac{1}{2}m{v_f}^2 + mgh_f = \frac{1}{2}m{v_i}^2 + mgh_i$$

Conservation of Mechanical Energy

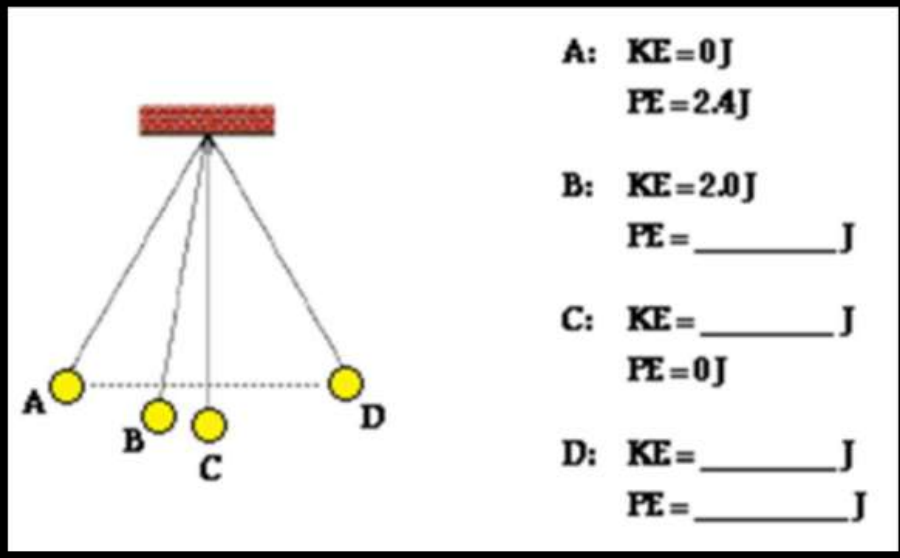




As the pendulum moves downward, its PE decreases while its KE increases. However, when it moves upward, its PE increases but its KE decreases. In either cases, the sum of PE and KE (i.e. ME) remains constant.

Application 6

Find the missing energies for the pendulum.



Find the missing energies for the pendulum.

Solution

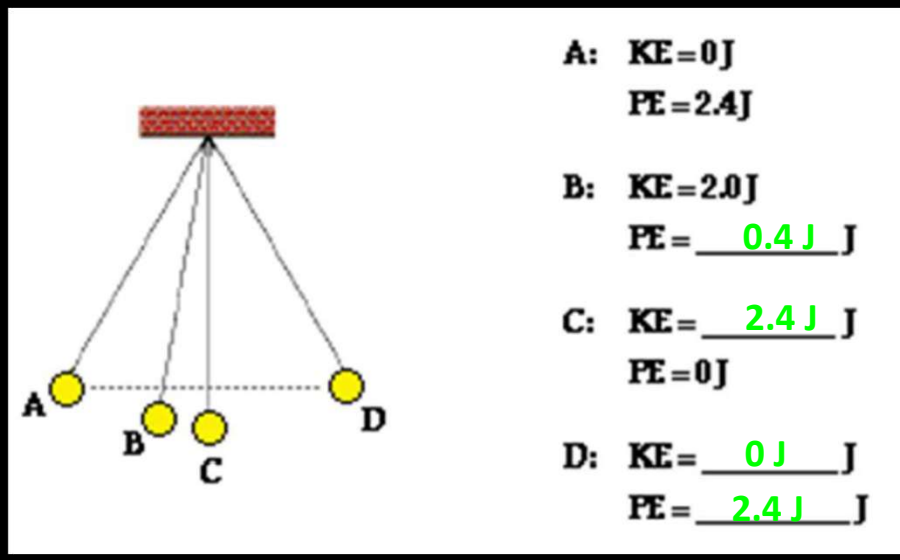
ME is constant then at every position which means (ME = KE + GPE) is the same.

Or at A: ME = KE+GPE = 0 + 2.4 = 2.4 J.
Then:

At B: GPE = 2.4 - 2 = 0.4 J

At C: KE = 2.4 - 0 = 2.4 J

At D: KE = 0 & GPE = 2.4 J (D is similar to A since they are at same level)



Application 7

A 5kg mass hanging 10 m high from a rope. If the rope is cut, at what speed does the mass hit the ground?

Solution

By the conservation of mechanical energy

$$KE_{bottom} + \cancelto{0}{GPE_{bottom}} = \cancelto{0}{KE_{top}} + GPE_{top}$$

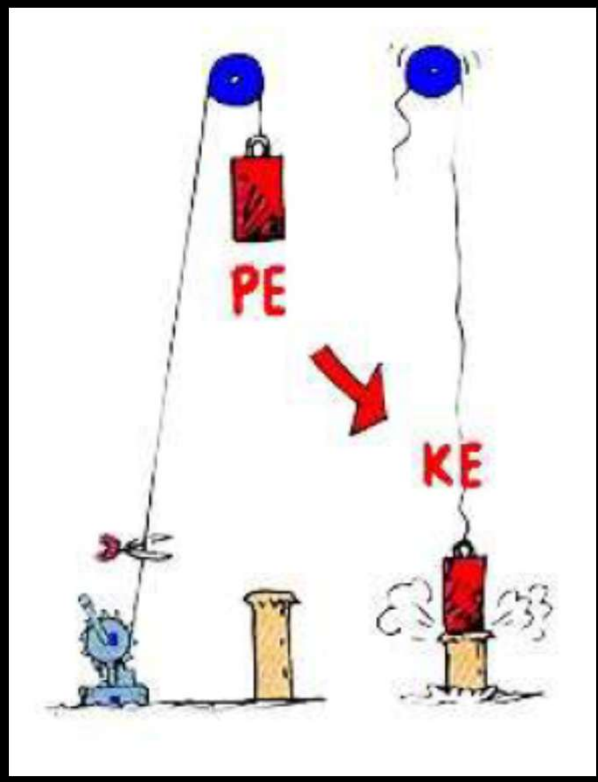
$$\frac{1}{2}m(v_{bottom})^2 = mgh_{top}$$

$$\frac{1}{2} \times 5 \times v^2 = 5 \times 10 \times 10$$

$$2.5 \times v^2 = 500$$

$$v^2 = \frac{500}{2.5} = 200$$

$$v = \sqrt{200} = 14.1\ m/s$$



Exercises

11. Example: A tennis ball thrown from ground level vertically upward at an initial speed of 10 m/s. Use the conservation of ME to find the maximum height it can reach.

12. Example: Starting from rest, a child zooms down a frictionless slide from an initial height of 3 m. What is her speed at the bottom of the slide? Assume the child has a mass of 25 kg.

13. Example: A pendulum bob is released from some initial height such that the speed of the bob at the bottom of the swing is 1.9 m/s. What is the initial height of the bob?

Power

- The rate at which work is done or the rate at which energy is transferred is called power:

$$Power = \frac{Work\ done}{Time\ taken} \quad \text{or} \quad P = \frac{w}{\Delta t} \quad \text{or} \quad P = \frac{E}{\Delta t}$$

- Since work is measured in joules (J) and time is seconds (s) then power is measured in (J/s) which is abbreviated watts (W).

Application 8

An electric light bulb converts 100 J of electric energy into light and heat every 1min. Find its power.

Solution

$\Delta t = 1 \text{ min} = 60 \text{ s}$

$P = E/\Delta t$

$= (100 \text{ J})/(60 \text{ s})$

$= 1.7 \text{ W}$

Exercises

16. Example: How long does it take a 19 kW steam engine to do 6.8×10^7 J of work?

17. Example: A 50 kg student climbs 5 m up a rope at a constant speed in 12.3 s.

a) How much work does the student do?

b) Find the student's power output.

Time equations

URM $\Rightarrow$

$$\begin{cases} a = 0 \\ v = constant = v_0 \\ x = v \cdot t + x_0 \end{cases}$$

UVRM $\Rightarrow$ $\begin{cases} a = constant \\ v = a \cdot t + v_0 \\ x = \frac{1}{2} a \cdot t^2 + V_0 \cdot t + x_0 \end{cases}$

$$If: \vec{a} \cdot \vec{v} > 0 \Rightarrow movement\ is\ accelerated$$

$$If: \vec{a} \cdot \vec{v} < 0 \Rightarrow movement\ is\ decelerated$$

SI units

Force strength in newtons (N), distance and linear abscissa, x, in meters (m), work and energy in joules (J), linear speed in meters per second (m/s), linear acceleration in meters per second squared (m/s^2).